

Flooding Analysis

Data Analysis STAT 4620/5620 Winter 24-25

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https://github.com/KyleA2001/STAT4620_5620-Flooding_Project.git

Abstract

Summary in 150 words (Best done last, summarizes the whole project)

Keywords

Listing out 10 key words max

Introduction

Over the past few years, Nova Scotia has faced various floods as a result of thunderstorms, heavy rain, and extreme weather conditions. These events severely impacted communities, leading to evacuations, property damage, infrastructure loss, and loss of life. Certain areas in Nova Scotia, particularly its coastal regions, are especially vulnerable to these events. With the increasing frequency and severity of extreme weather due to climate change, the risk of flooding in Nova Scotia is expected to grow.

Modeling these rare extreme events is challenging due to their rarity, yet their societal and economic impacts can be profound and severe. This research aims to understand the relationships between various climatic factors and storm surge levels at selected locations across Nova Scotia. The analysis will be followed by predictions of the return periods and surge levels in the studied regions. Sophisticated and relevant statistical models will be employed to investigate this research.

This research will address two key questions:

1. What is the relationship between various climatic factors and storm surge levels across Nova Scotia?
2. How can the most significant correlated factors be used to predict future flooding catastrophes in Nova Scotia?

Through these investigations, this study aims to contribute valuable predictions and insights that will aid in understanding and reducing future flooding risks in the province.

Data Description

In this section, we describe the data used for the analysis. Below is the summary table for the data:

Variables	Units	Description	Frequency
Date	days	Date when data was recorded	Daily
Location		Halifax and Yarmouth	
Water Level	m (meters)	Height above datum or reference system	Hourly
Storm Surge	m (meters)	Water rise due to water level and tide height	Hourly
Air Temperature	°C	Air temperature	Daily min/max, Hourly avg
Wind Speed	km/h	Wind speed	Daily min/max, Hourly avg
Rainfall	mm	Precipitation (not including snow)	Real daily value

Table 1: Data description summary

Methods

– Provide a brief description of the analytical tools you will used – Discuss why the method is appropriate in the context of the research question and the data, including any assumptions required by the method.

During the initial start of the project, a various amount of methods were brought up. It was decided that the two main methods to answer question 1 of the research would be GLMs and Time Series. GLMs were considered because it would help provided a baseline to figure out what factors affect tidal storm surge. Unfortunately, there were some limitations and violated assumptions that prevented the use of this model.

Generalized Linear Models (GLMs) are a traditional linear regression that handles a wider range of data types (Continuous, count, etc...). What this model does not account for is the limitations of time-dependent data. The data collected in this research depends on the time it was collected, to find a relationship with storm surge and use for future prediction. Without the use of time data in the model, their will not be any accurate predictions.

An assumption that GLMs have is that all observations are independent. However, the data collected violates this assumption because the observations are dependent on time, meaning that values recorded at one time point are likely influenced by previous observations. This temporal dependency introduces autocorrelation, where data points closer in time are more similar, making GLMs less suitable for capturing these patterns. As a result, applying a method that accounts for time dependence was necessary to better model the data.

The dependence on time lead to the use of GAMMs (Generalized Additive Mixed Models). “Generalized Additive Models (GAMs) are a versatile statistical modeling technique used to analyze complex relationships within data. Unlike linear models, GAMs can capture non-linear patterns by combining multiple smooth functions of predictor variables.” (Basheer, 2024). The model does not account for time dependence or correlated observations. This is where the model choice extends to GAMMs that accounts for both.

GAMMs assume that the random effects are independent. Since the data collected is from two different locations, the random effect from each location should be independent from each other. Another assumption is that Storm Surge should be able to follow a certain distribution.

For the research the assumption is that the model follows a Normal Distribution. One of the important assumptions is temporal correlation, where observations in the past influence future values. The random effect to the model accounts for this.

(Insert code for acf)

ACF (Autocorrelation Function) is a method that will show how much a variable at one time is related to its value at previous times. Since the research is studying storm surge, ACF will show if today's storm surge value was influenced from its values yesterday, before than, etc... ACF helps identity any patterns in the time series data, like seasonality or trends. If the current and past value show to be highly correlated, then the data is not random and there is some predictability in the structure.

Using ACF on the GAMM model, there is high correlation between the data points, suggesting that the data collected for each time are dependent. This violates the assumption of GAMMs, and therefore makes it unusable for the model.

The other model that will be used for this analysis is Time Series. Time Series models the sequence of data points in an equally spaced time interval. The method is used to analyze trends, patterns, and behaviours over time. The importance of this model is to analyze historical data to help understand any underlying patterns, predict future values, and see how (or if any for our research) certain factors influence storm surge over time.

The goal of the research is to see what climatic factors affect storm surge over time. Time series modeling will help examine if these variables have a time-based relationship with storm surge. GAMMs was helpful for modeling non-linear, time-dependent relationships between the predictors and the response variable. However, it might not accurately represent the complex nature of time series data.

Time Series models assume that the data is stationary and no autocorrelation in the residuals. (Unsure what else to add)

Since the research analyzes storm surge data, ACF will help look at temporal dependencies. This will help identify how much past climatic factors influence storm surge behaviour.

CCF(Cross-Correlation Function) is the same as ACF, but instead examines the relationship between two different time series, with more flexibility to look at different points in time. When using CCF for the model, it is testing how strongly the response variable (Storm Surge) is related to another variable (predictor variables like rain, wind speed, and temperature) over different lags. For the research, CCF can help to see if rain, wind speed, and/or temperature have an immediate or delayed effect on storm surge.

VAR(Vector Auto regression) (not 100% about this model)

Granger Causality (Not 100% sure)

Question 2's method is missing...

Analysis

- Show the code used to actually apply the methods to the data.
- Include model validation.

Results

- Present and discuss the outputs of your methods.
- Include any visualisations of the analysis.

Conclusions

- Discuss the results of the previous section in the context of the research question.
- Give an answer to your research question and discuss any uncertainty in your results.
- If appropriate, discuss why the research question cannot be fully answered using the data and techniques available.

References

- Cite data sources.
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Cite literature and other sources useful for explaining the background information, research question, and analytical tools.

<https://www.analyticsvidhya.com/blog/2023/09/understanding-generalized-additive-models-gams-a-comprehensive-guide/#h-definition-of-gams>, Cite later